

Scleral Necrosis after Plaque Radiotherapy of Uveal Melanoma: A Case-Control Study

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Purpose: To identify risk factors and outcome of scleral necrosis after plaque radiotherapy of uveal melanoma.

Design: Case-control study.

Participants: A total of 73 cases with scleral necrosis and 73 controls without necrosis after plaque radiotherapy. Controls were matched for anteroposterior tumor epicenter and follow-up duration.

Intervention: Plaque radiotherapy with iodine-125, cobalt-60, iridium-192, or ruthenium-106.

Main Outcome Measures: Scleral necrosis.

Results: Of 5057 patients treated with plaque radiotherapy for uveal melanoma, 73 (1%) developed radiotherapy-induced scleral necrosis. Scleral necrosis occurred in <1% of patients (3/1140) when plaque radiotherapy was used for tumors <3 mm in thickness, 1% of patients (33/3155) with 3- to 8-mm tumor thickness, and 5% of patients (37/762) with >8-mm-thick tumors. On the basis of tumor location, scleral necrosis was detected after plaque radiotherapy of iris melanoma in 0% of patients (0/91), ciliary body melanoma in 29% of patients (67/235), and choroid melanoma in <1% of patients (6/4731). The mean time interval between plaque radiotherapy and scleral necrosis was 32 months (median, 23 months; range, 4–126 months). The mean basal dimension of scleral necrosis was 4 mm (median, 3 mm; range, 1–15 mm), equivalent to 29% of mean tumor base (median, 24%; range, 6%–100%) and 22% of mean plaque size (median, 19%; range, 5%–75%). Multivariate analysis of factors that predicted clinically evident scleral necrosis included ciliary body ($P = 0.0001$) and pars plana to ora serrata ($P < 0.0001$) locations of anterior tumor margin, tumor thickness ≥ 6 mm ($P = 0.0001$), and radiation dose ≥ 400 Gy to the outer sclera ($P = 0.0455$). Scleral necrosis remained stable in 48% of patients (35/73), increased in size/severity in 48% of patients (35/73), or progressed to scleral perforation in 4% of patients (3/73) over a mean follow-up of 79 months (median, 54 months; range, 5–351 months). Treatment of scleral necrosis included observation in 81% of patients (59/73), scleral patch graft in 14% of patients (10/73), and enucleation in 5% of patients (4/73).

Conclusions: Scleral necrosis after plaque radiotherapy of uveal melanoma was detected in 1% of cases. Factors predictive of scleral necrosis included increasing tumor thickness, ciliary body and peripheral choroidal location, and higher radiation dose to sclera. Most patients (81%) did not require treatment, and 4% evolved to full-thickness perforation.

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The management of uveal melanoma depends on tumor size and location and includes focal transpupillary thermotherapy for small borderline melanoma; radiotherapy (charged particle irradiation or plaque radiotherapy) for small, medium, and large melanoma; resection for anterior tumors; and enucleation for large melanoma or those encircling the optic disc.¹ Plaque radiotherapy is the most commonly used method because of the advantages of high local tumor control and preservation of the eye with some vision.^{1–3} The Collaborative Ocular Melanoma Study revealed no difference in survival outcomes and little difference in quality of life outcomes comparing patients treated with plaque radiotherapy with those treated with enucleation, further favoring the conservative use of plaque radiotherapy.^{4–7}

The first description of focal ocular radiotherapy for uveal melanoma was by Moore⁸ in 1930, when he directly

inserted radon seeds into the thickest portion of uveal melanoma, which resulted in reduction of the mass to one quarter of the original tumor size. Since then, various radioactive isotopes have been used, including cobalt-60 (⁶⁰Co), iridium-192 (¹⁹²Ir), ruthenium-106 (¹⁰⁶Ru), gold-198 (¹⁹⁸Au), palladium-103 (¹⁰³Pd), and iodine-125 (¹²⁵I).^{9–15} ¹²⁵I plaque radiotherapy is the most commonly used radionuclide in the United States. We currently use only ¹²⁵I radioisotope at the Ocular Oncology Service, Wills Eye Institute. Ocular complications after plaque radiotherapy include radiation-induced dry eye in 8% of patients,¹⁶ diplopia in 10% of patients,¹⁷ strabismus in 2% of patients,¹⁸ keratitis in 4% to 21% of patients,^{16,19} iris neovascularization in 4% to 23% of patients,^{20,21} neovascular glaucoma in 2% to 45% of patients,^{20,22} cataract in 8% to 68% of patients,^{20,23} vitreous hemorrhage in 4% to 18% of patients,²⁴